$$\mathbf{Q}_t^1 = \mathbf{Q}_t^0 + \int_0^1 \left[v_{\theta_Q}^Q\Big(\mathbf{Q}_t^\tau, \tau; \mathcal{K}_t^Q(\tau)\Big) + \lambda_\tau^Q \left(\mathbf{J}_t^Q(\tau)\right)^\mathsf{T} \mathbf{P}^Q \mathbf{W}_t^Q(\tau) \left(\mathbf{Y}_t^Q(\tau) - \hat{\mathbf{Q}}_t^1(\tau)\right)\right] \mathrm{d}\tau, \qquad Q \in \{Z, A, X\}$$